

Object Detection

Acknowledgement: Most content of this lecture are based on Deep Learning for Computer Vision by Prof. Vineeth

Classification vs Localization vs Detection

Classification



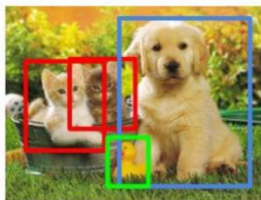
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

**Instance
Segmentation**



CAT, DOG, DUCK

Single object

Multiple objects

Face Detection

Face Detection

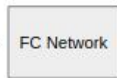
- It is also object detection
- Pre-Deep Learning Era: Viola-Jones Algorithm

Object Detection using CNNs

Training Step:

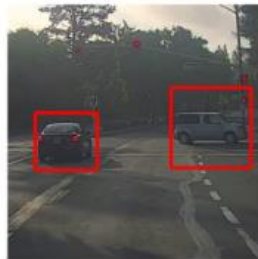


Testing Step:

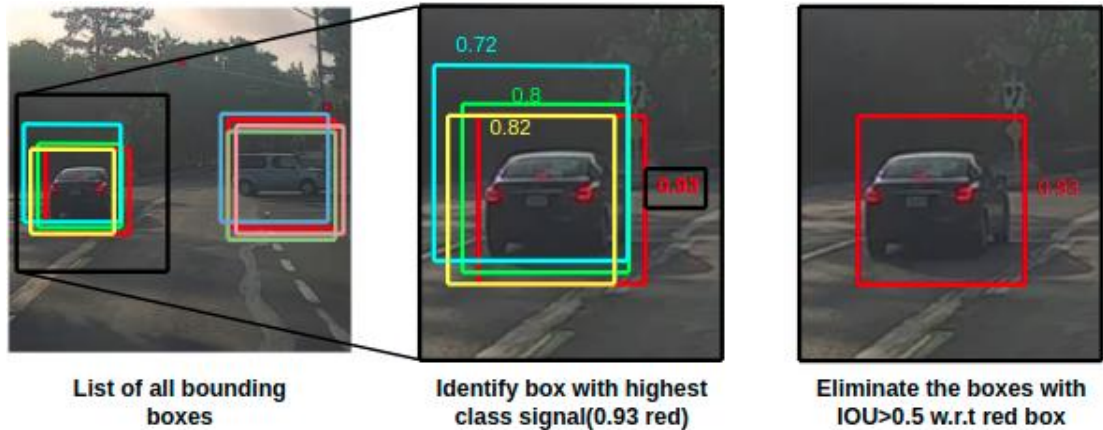


Car/No Car

Sliding windows
of multiple
scales



Non-Maximum Suppression using Class Scores

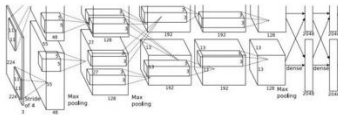


CNN as feature extractor



CNN as feature extractor

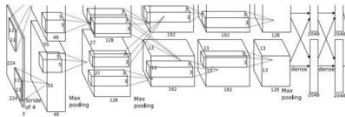
Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



Dog? NO
Cat? NO
Background? YES

CNN as feature extractor

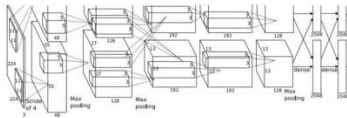
Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



Dog? YES
Cat? NO
Background? NO

CNN as feature extractor

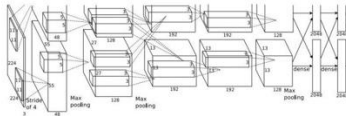
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CNN as feature extractor

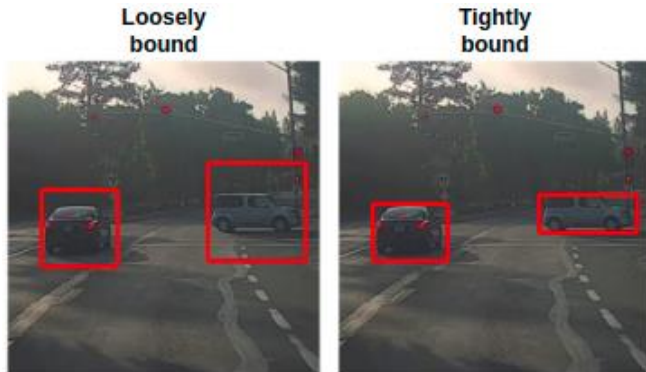
Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



Dog? NO
Cat? YES
Background? NO

Object Detection using CNNs and Sliding Windows: Disadvantages

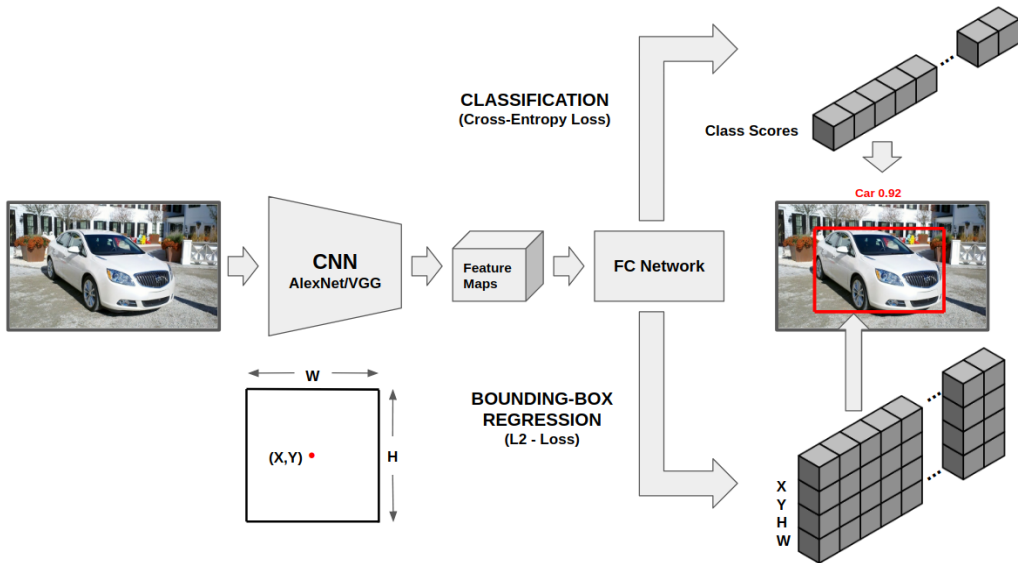
- Bounding boxes may not be tightly around the object
- Evaluating a CNN network on all sliding windows is computationally intensive and time consuming



CNN as feature extractor

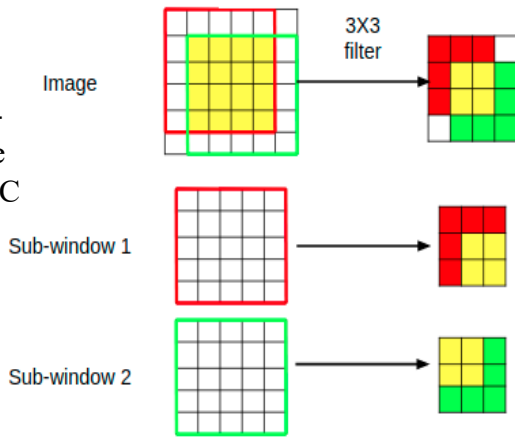
- What could be the problems?
 - Suppose we have a 600 x 600 image, if sliding window size is 20 x 20, then have $(600-20+1) \times (600-20+1) = \sim 330,000$ windows
 - Sometimes we want to have more accurate results -> multi-scale detection
 - Resize image
 - Multi-scale sliding window
- For each image, we need to do the forward pass in the CNN for $\sim 330,000$ times. -> Slow!!!

Object Localization using Bounding Box Regression



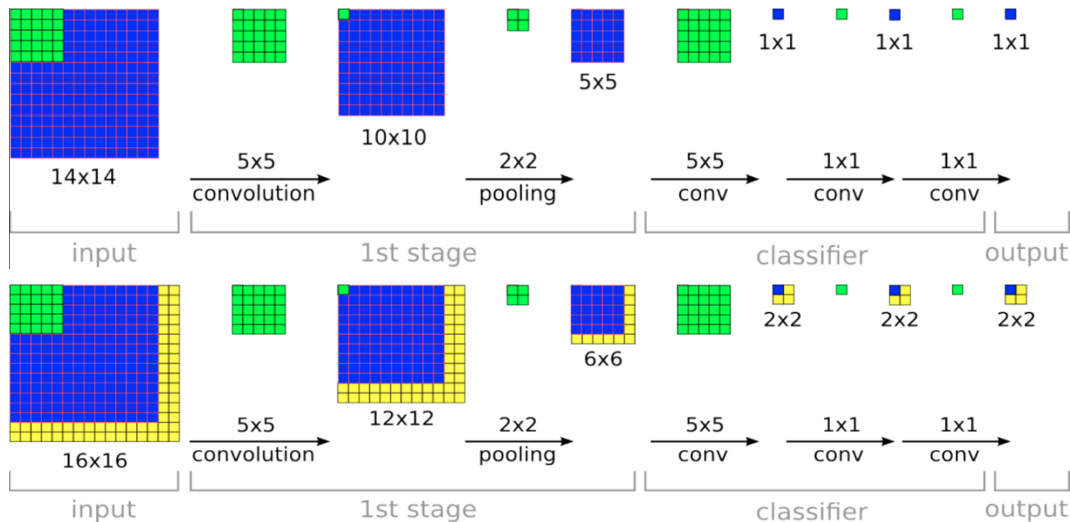
OverFeat: Integrated Recognition, Localization and Detection

- Winner of ILSVRC 2013 Localization Challenge
- **Intuition:** Avoid computation time over sub-windows by applying filter directly to image
- Used only Convolutional layers instead of FC



- Sermanet et al, OverFeat: Integrated Recognition, Localization and Detection using Convolutional Networks, ICLR 2014

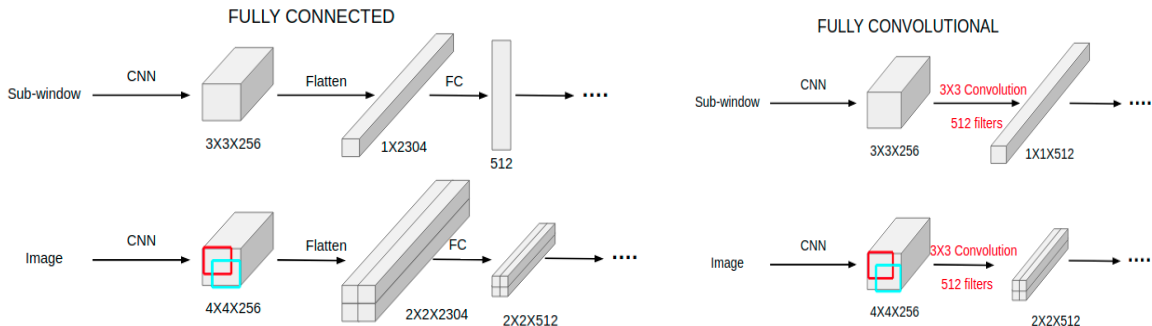
OverFeat: Integrated Recognition, Localization and Detection



- Sermanet et al, OverFeat: Integrated Recognition, Localization and Detection using Convolutional Networks, ICLR 2014

OverFeat: Integrated Recognition, Localization and Detection

- The network used is fully convolutional, replacing fully connected layers with convolutional layers. Why?



- Sermanet et al, OverFeat: Integrated Recognition, Localization and Detection using Convolutional Networks, ICLR 2014

Contemporary Object Detection Methods

Region Proposal-based:

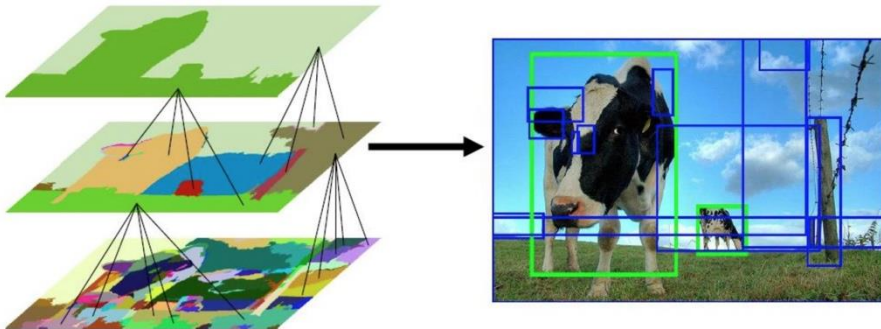
- Two-stage detection framework
- In the first stage, potential object regions are proposed (through methods such as Selective Search or Region Proposal Network)
- In the second stage, a classifier processes the candidate regions
- More robust in performance but slower

Dense Sampling-based:

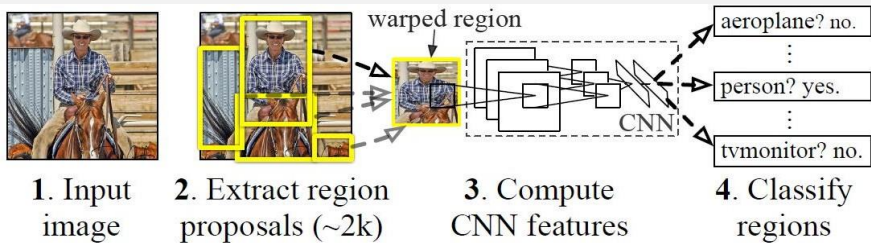
- One-stage detection framework
- Integrates region proposals and detection by acting on a dense sampling of possible locations
- Simple and fast but performance not as good as Region Proposal-based methods

Region Proposal

- Use some fast algorithms to filter out some regions first, only feed the potential region (region proposals) into CNN
- E.g. selective search



R-CNN (Girshick et al. CVPR 2014)



- Replace sliding windows with “selective search” region proposals (Uijilings et al. IJCV 2013)
- Extract rectangles around regions and resize to 227x227
- Extract features with fine-tuned CNN (that was initialized with network trained on ImageNet before training)
- Classify last layer of network features with SVM, refine bounding box localization (bbox regression) simultaneously

Bounding Box Regression

- Intuition

- If you observe part of the object, according to the seen examples, you should be able to refine the localization
- E.g. given the red box below, since you've seen many airplanes, you know this is not a good localization, you will adjust it to the green one



Bounding Box Regression

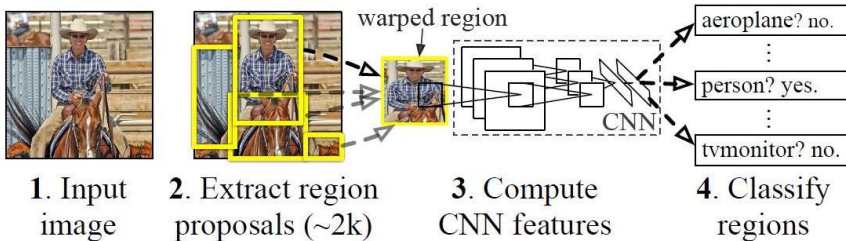
- Intuition

- If you observe part of the object, according to the seen examples, you should be able to refine the localization
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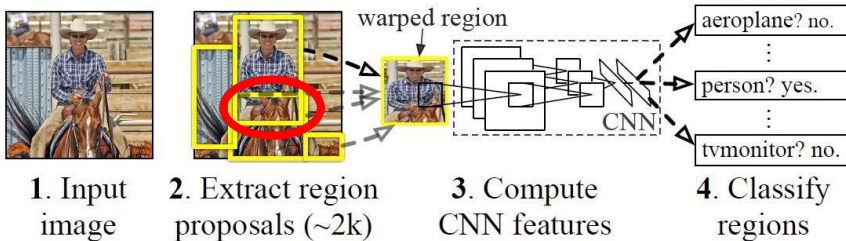
R-CNN (Girshick et al. CVPR 2014)

- What could be the problems?



R-CNN (Girshick et al. CVPR 2014)

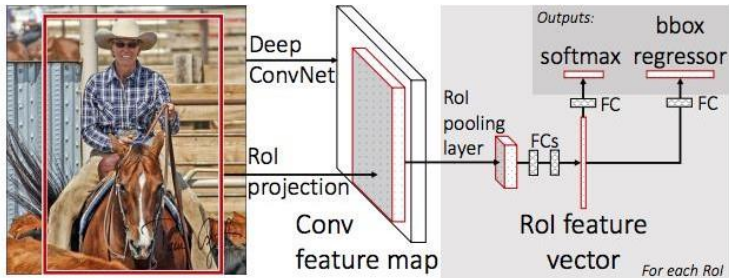
- What could be the problems?
 - Repetitive computation! For overlapping regions, we feed it multiple times into CNN



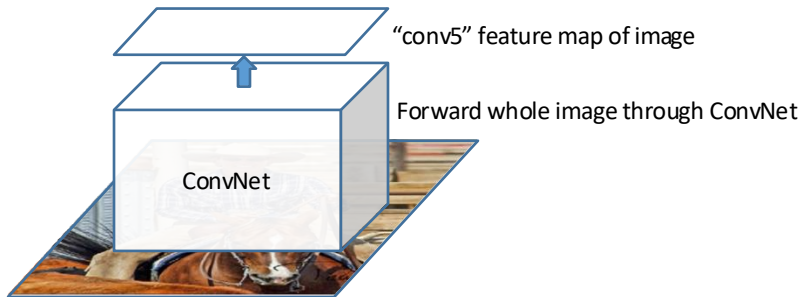
Fast R-CNN (Girshick ICCV 2015)

- Solution

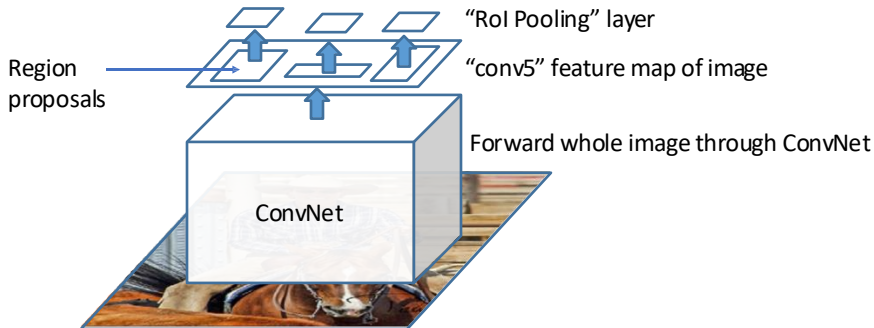
- Why not feed the whole image into CNN only once! Then crop features instead of image itself



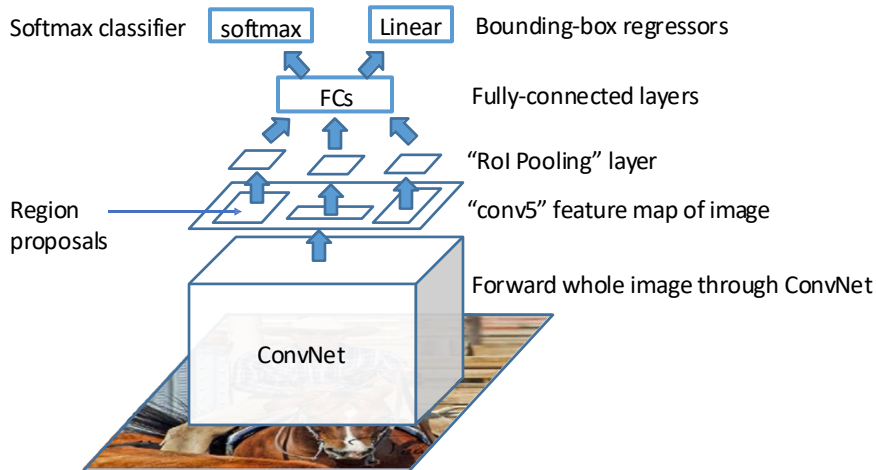
Fast R-CNN (Girshick ICCV 2015)



Fast R-CNN (Girshick ICCV 2015)

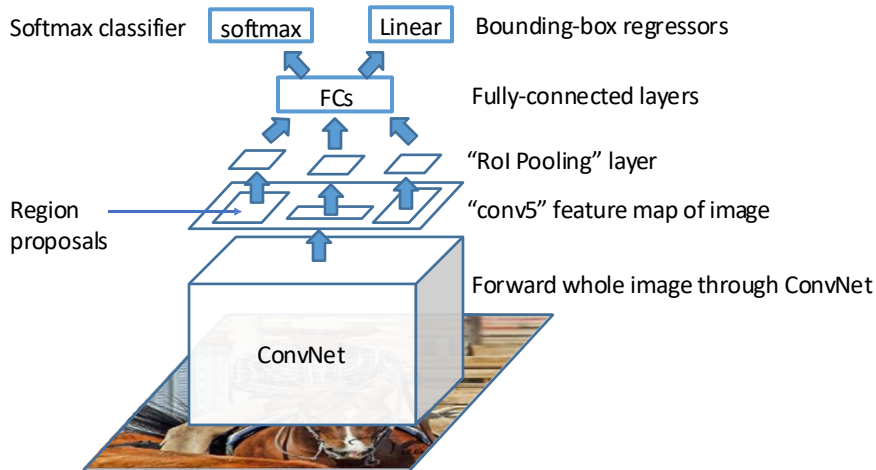


Fast R-CNN (Girshick ICCV 2015)

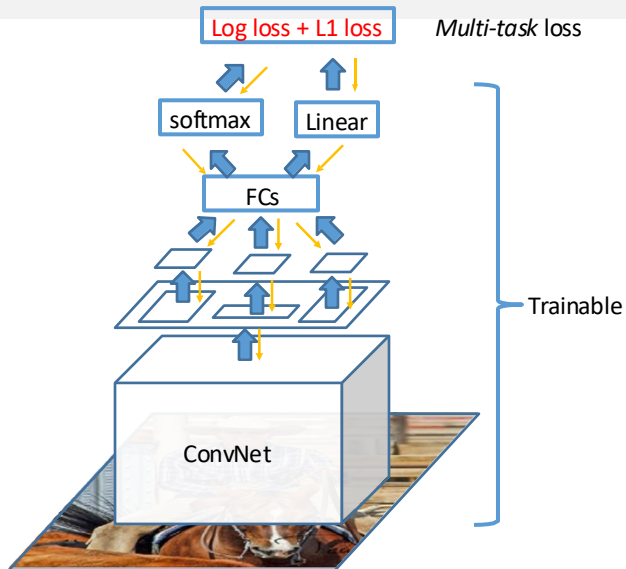


Fast R-CNN (Girshick ICCV 2015)

Rather than using post-hoc bounding-box regressors, bounding-box regression is implemented as an additional linear layer in the network



Fast R-CNN (Girshick ICCV 2015)

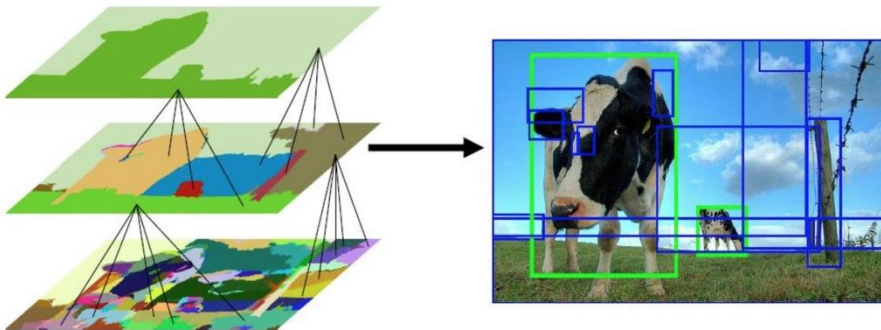


Fast R-CNN (Girshick ICCV 2015)

- What could be the problems?

Fast R-CNN (Girshick ICCV 2015)

- What could be the problems?
 - Why we need the region proposal pre-processing step? That's not “deep learning” at all. Not cool!



Faster R-CNN (Ren et al. NIPS 2015)

- Solution

- Why not generate region proposals using CNN??!

-> RPN

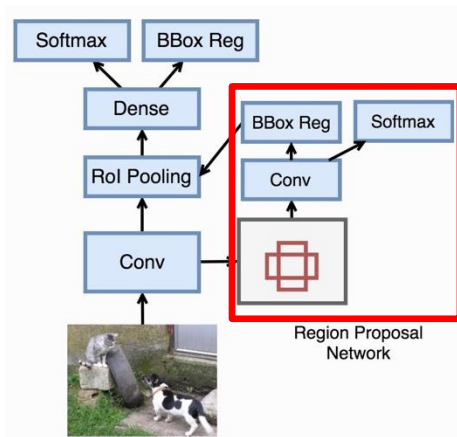


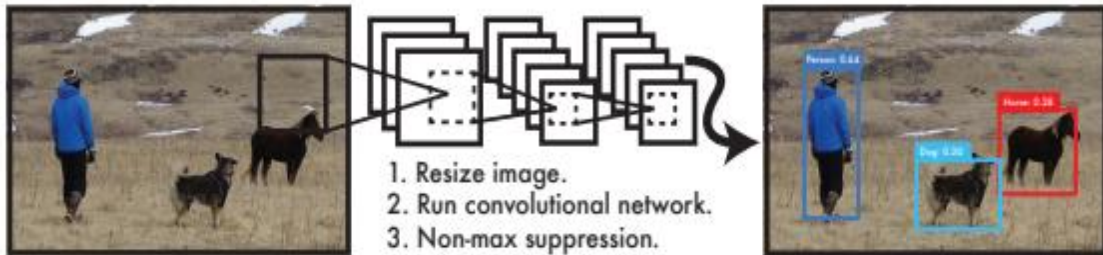
Image credit:

<https://arxiv.org/pdf/1506.01497.pdf>

http://zh.gluon.ai/chapter_computer-vision/object-detection.html

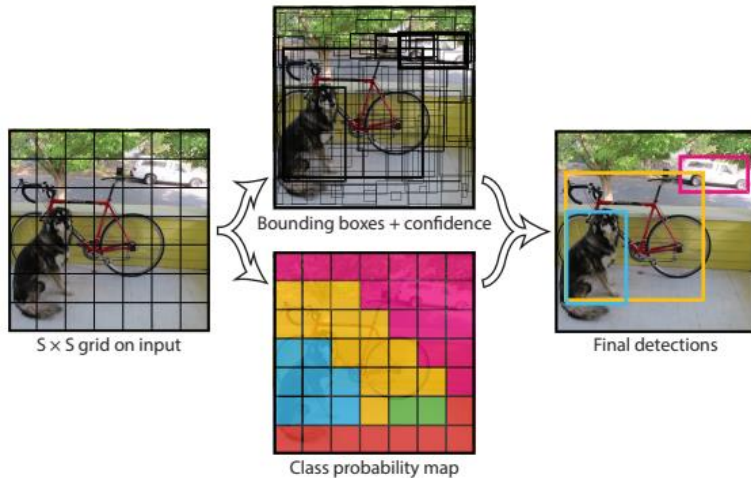
You Only Look Once (YOLO): v1

- Single-stage detector based on Overfeat
- Speed with good performance the main aim

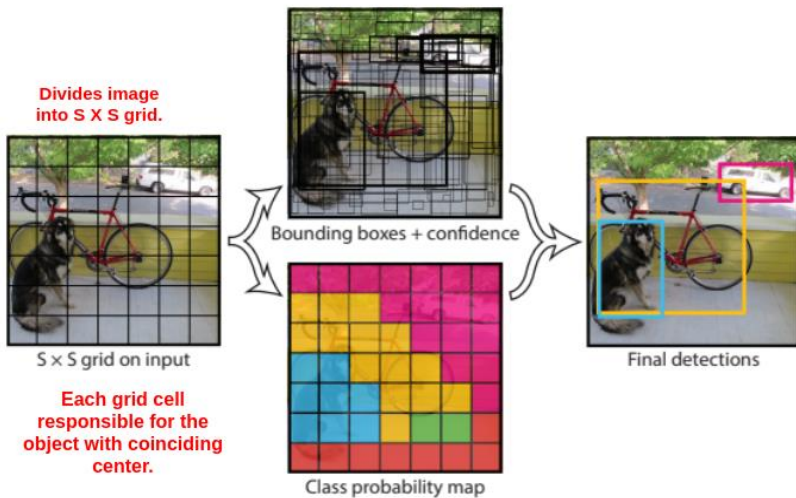


- Redmon et al, You Only Look Once: Unied, Real-Time Object Detection, CVPR 2016

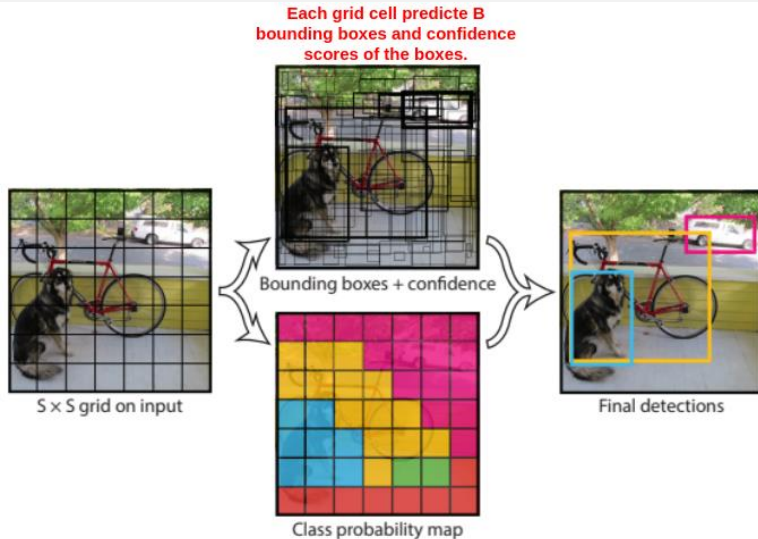
YOLO v1: Unified Detection



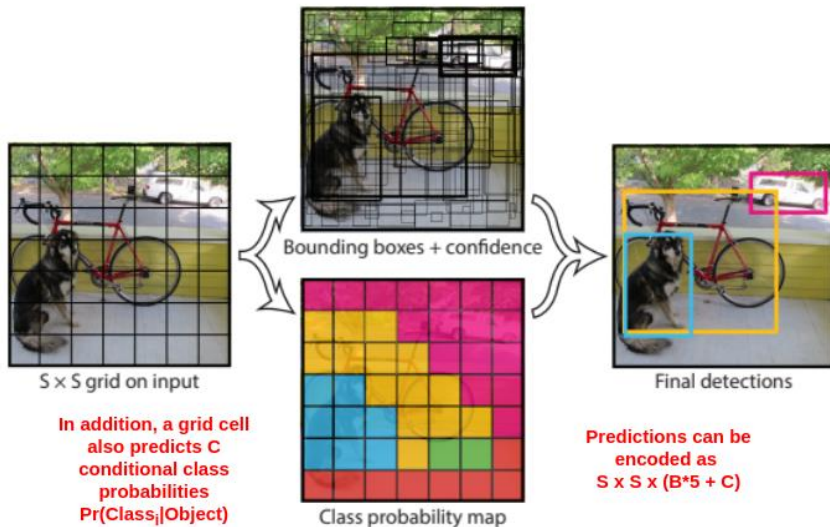
YOLO v1: Unified Detection



YOLO v1: Unified Detection



YOLO v1: Unified Detection



YOLO v1: Bounding Boxes and Confidence Scores

Bounding Boxes:

- Each bounding box gives 4 coordinates x, y, w, h
- (x, y) = Coordinates representing center of the object relative to the grid cell
- (w, h) = Width and height of the object relative to the whole image

Confidence Score:

- Reflects how confident the model is that the box contains an object and also how accurate it thinks the box is
- Formally, $\text{Confidence} = Pr(\text{Object}) * IOU_{pred}^{truth}$

Conditional Class Probabilities:

- Regardless of number of boxes B, we only predict one set of class probabilities per grid cell
- At test time, class-specific confidence scores for each box are given by:

$$Pr(Class_i | Object) * Pr(Object) * IOU_{pred}^{truth} = Pr(Class_i) * IOU_{pred}^{truth}$$

Loss Function

Loss function used to train YOLO v1 given by:

$$\begin{aligned} & \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} [(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2] \\ & + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(\sqrt{w_i} - \sqrt{\hat{w}_i})^2 + (\sqrt{h_i} - \sqrt{\hat{h}_i})^2 \right] + \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\ & + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 + \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2 \end{aligned}$$

where:

- $\mathbb{1}_i^{\text{obj}}$ denotes if object appears in cell i
- $\mathbb{1}_{ij}^{\text{obj}}$ denotes that j^{th} bounding box predictor in cell i is 'responsible' for that prediction

Yolo Limitations

- Detects only a small number of objects
- Cannot detect objects that are small or close due to strong spatial constraints
- High localization error
- Relatively low recall

References

- Please refer to Mitesh Khapra's original lecture slides (and videos) for more detailed explanation of some of these topics, available at [CS7015: Deep Learning](#).
- Also refer to Vineet Balasubramanian original lecture slide and videos for more detailed information [CS5370: Deep Learning for Vision](#)

Other good resources:

[Deep Learning Book:](#)

Stanford [CS231n Notes](#)